

# Force Sensorless Admittance Control for Physical Human–Robot Interaction with Direct-Drive Manipulators

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**Abstract**—To enhance the compliance of direct-drive manipulators in physical human–robot interaction (pHRI), this paper introduces a force-sensorless admittance control method. First, a generalized momentum observer (GMO) is adopted to estimate the external interaction force, which is then transformed into an end-effector displacement correction through an admittance control algorithm. Then, a radial basis function neural network (RBFNN) is introduced to offset unmodeled dynamics, ensuring accurate tracking of the corrected reference trajectory. Finally, simulations on a 6 degree of freedom (DOF) direct-drive manipulator demonstrate that the proposed method enables precise trajectory tracking and responsive adaptation to human movements, thereby verifying its effectiveness and feasibility.

**Index Terms**—direct-drive manipulator, physical human–robot interaction (pHRI), radial basis function neural network (RBFNN), admittance control.

## I. INTRODUCTION

In recent years, robotic systems have found rapidly expanding applications in manufacturing, industrial production, and logistics. This growth is driven by advances in enabling technologies such as high-resolution perception, precise positioning, and environment-aware motion planning algorithms [1], [2]. In such domains, robots primarily accomplish tasks through accurate control of end-effector positions. By contrast, in scenarios involving physical human–robot interaction (pHRI), such as medical rehabilitation and service industries, robots must not only execute tasks with high precision but also interpret and respond to human motion intentions in real time. These capabilities are essential for both task performance and operational safety, as they mitigate potential hazards from

excessive or unexpected interaction forces [3]. Consequently, collaborative pHRI operations impose stricter requirements on the force-control accuracy and dynamic responsiveness of robotic manipulators, motivating the development of advanced control strategies to ensure safe and compliant interaction.

Most robotic manipulators in practical applications still rely on indirect-drive systems, in which motion is transmitted through mechanical elements such as gears and reducers. While widely used, these mechanisms are inherently prone to assembly errors, backlash, and elastic deformation [4]. Such imperfections introduce frictional losses, mechanical noise, accelerated component wear, and positioning errors, thereby increasing operational and maintenance costs. By contrast, direct-drive technology establishes a direct connection between the motor rotor and the manipulator joint, eliminating intermediate transmission elements [5]. This design removes errors caused by gear backlash and transmission-chain elasticity, thereby greatly improving joint positioning accuracy and consistency [6]. Furthermore, the reduction of vulnerable components decreases maintenance demands, enhances operational efficiency, and extends service life [7]. Owing to these advantages, direct-drive manipulators are particularly well suited for applications that demand high safety, compliance, and precision, especially in pHRI scenarios.

To address compliance challenges in pHRI, a variety of control approaches have been devised. The earliest approach, position/force hybrid control [8], enabled compliant interaction by simultaneously regulating end-effector position and contact force. This method is later extended and applied to pHRI scenarios [9]. The impedance control approach was first presented by Hogan, which builds a relationship between contact forces and system states based on an impedance model [10]. In contrast to position/force hybrid control, impedance

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control exhibits greater robustness, as it eliminates the need for switching between contact and non-contact modes. Based on causality, impedance control is typically classified into impedance and admittance control [11]. In recent years, these two approaches have become the most widely used strategies in pHRI. As an illustration, researchers in [12] developed a learning-based impedance controller designed to enable robotic systems to track a predefined impedance model and accomplish interaction objectives. A unified admittance-based controller was designed in [13] to realize pHRI in constrained task spaces. In [14], researchers developed an admittance-based controller to accomplish pHRI within the Constrained Task Domain.

Although the above proposed admittance control can improve pHRI performance, it typically requires access to external interaction forces. However, torque sensors are expensive, and many manipulators are not equipped with built-in force sensors, which limits their application. Direct-drive manipulators eliminate traditional transmission components such as reducers by directly connecting the motor rotor to the load. In this design, the external interaction force applied at the joint load side acts directly on the motor, enabling sensorless pHRI and significantly reducing production costs. Nevertheless, removing the transmission components also introduces new control challenges. Characterized as a time-varying, coupled, multi-input–multi-output nonlinear system, the joint dynamics of manipulators are impacted by the motion of adjacent joints. In direct-drive manipulators, unmodeled dynamics act directly on the motor, amplifying disturbances, increasing control difficulty, and severely affecting positioning accuracy [15]. Therefore, existing admittance control methods cannot be directly applied to direct-drive manipulators for pHRI and must be adapted to address their specific characteristics.

To enable direct-drive manipulators with force-sensorless pHRI, this paper proposes an admittance control method to enhance interaction compliance. The remainder of this paper is structured as follows: Section II sets forth the dynamic model of the direct-drive manipulator and defines the research problem. The sensorless admittance control approach proposed in this study is introduced in Section III. Section IV reports simulation studies on a six-degree-of-freedom (DOF) direct-drive manipulator and discusses the corresponding results. Ultimately, Section V draws conclusions for the paper and outlines potential future research directions.

## II. PREREQUISITES AND PROBLEM DESCRIPTION

### A. Dynamic Model of the Manipulator System

As shown in Fig. 1, the schematic of the serial manipulator is provided, with  $m_i$ ,  $q_i$ , and  $c_i$  representing the mass, rotation angle, and center of mass of the  $i$ -th revolute joint.

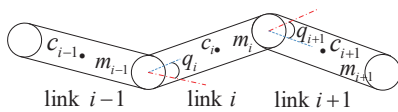


Fig. 1. Structure diagram of serial manipulator

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) + \tau_f(\dot{q}) = \tau + \tau_{ext} \quad (1)$$

where  $M(q) \in \mathbb{R}^{n \times n}$  denotes the inertia matrix, while  $C(q, \dot{q}) \in \mathbb{R}^{n \times n}$  is the Coriolis torque matrix. The gravity torque is expressed by  $G(q) \in \mathbb{R}^n$ , and joint friction is modeled through  $\tau_f(\dot{q})$ . The vectors  $q, \dot{q}, \ddot{q} \in \mathbb{R}^n$  correspond to the joint positions, velocities, and accelerations, respectively. In addition,  $\tau \in \mathbb{R}^n$  stands for the actuation torque, and  $\tau_{ext} \in \mathbb{R}^n$  denotes the external torque arising from human interaction or environmental contact.

Given that pHRI occurs within the task space, converting the manipulator from joint space to task space is indispensable. The forward kinematics is defined by  $\varphi(q)$ , which yields the end-effector position  $x \in \mathbb{R}^m$  from the joint vector  $q \in \mathbb{R}^n$ :

$$x = \varphi(q) \quad (2)$$

By differentiating (2) with respect to time, it can be expressed as follows :

$$\dot{x} = J(q)\dot{q} \quad (3)$$

where  $J(q) \in \mathbb{R}^{m \times n}$  stands for the Jacobian matrix corresponding to the manipulator, and  $\dot{x} \in \mathbb{R}^m$  denotes the velocities of the manipulator in the task space.

From (2) and (3),  $\dot{q}$  and  $\ddot{q}$  can be deduced as:

$$\dot{q} = J^+(q)\dot{x} \quad (4)$$

$$\ddot{q} = \dot{J}^+(q)\dot{x} + J^+(q)\ddot{x} \quad (5)$$

where  $\dot{J}^+$  is defined as the pseudoinverse of  $J(q)$ , while  $\ddot{x} \in \mathbb{R}^m$  stands for the end-effector acceleration in the task space acceleration.

When (4) and (5) are substituted into (1), this gives rise to the dynamic model of the manipulator in task space:

$$M_x(q)\ddot{x} + C_x(q, \dot{x})\dot{x} + G_x(q) + F_{fx} = F + F_{ext} \quad (6)$$

where

$$M_x(q) = J^{+\top}(q)M(q)J^+(q) \quad (7)$$

$$C_x(q, \dot{x}) = J^{+\top}(q) \left( C(q, \dot{q}) - M(q)J^+(q)\dot{J}^+(q) \right) J^+(q) \quad (8)$$

$$G_x(q) = J^{+\top}(q)G(q) \quad (9)$$

$$F_{fx} = J^{+\top}(q)\tau_f \quad (10)$$

$$F = J^{+\top}(q)\tau \quad (11)$$

$$F_{ext} = J^{+\top}(q)\tau_{ext} \quad (12)$$

where the superscript  $\top$  signifies the matrix transpose operation. In the task domain,  $M_x(q) \in \mathbb{R}^{m \times m}$ ,  $C_x(q, \dot{q}) \in \mathbb{R}^{m \times m}$ , and  $G_x(q) \in \mathbb{R}^m$  are associated with the inertia, Coriolis, and gravity components, respectively. The lumped disturbance is denoted by  $F_{fx} \in \mathbb{R}^m$ , and the control input by  $F \in \mathbb{R}^m$ . The external interaction force is given by  $F_{ext} \in \mathbb{R}^m$ , which accounts for both forces and torques applied between the manipulator end effector and the human or environment.

### B. The model of Direct-drive Manipulator

The motion transmission principle of the manipulator joint is presented in Fig. 2, where the effect of the load transmitted through the drive system on the motor can be expressed as:

$${}^m M_i = \frac{{}^l M_i}{R_i^2} \quad (13)$$

$${}^m G_i = \frac{{}^l G_i}{R_i} \quad (14)$$

where  $R_i$  is defined as the transmission ratio of the  $i$ th joint;  ${}^l M_i$  and  ${}^l G_i$  are the payload-side inertia and gravity, whereas  ${}^m M_i$  and  ${}^m G_i$  represent the motor-side inertia and gravity reflected through the transmission.

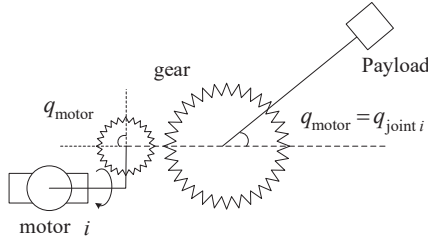


Fig. 2. Transmission mechanism schematic diagram

For direct-drive manipulators, the joint shaft is directly connected to the motor rotor, so the transmission ratio in (13) and (14) is  $R_i = 1$ . As a result, the transmission mechanism shown in Fig. 2 can be simplified to that in Fig. 3.

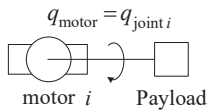


Fig. 3. Direct drive transmission schematic diagram

From equations (13) – (14), it can be seen that in indirectly driven manipulators, the effects of load-side inertia and gravitational torque on the motor are significantly attenuated by the reducer, making the impact of modeling errors on the motor side negligible. In contrast, direct-drive manipulators eliminate the transmission mechanism, and the joint is directly coupled to the motor rotor. Consequently, unmodeled dynamics act directly on the motor, intensifying the effects of variable inertia coupling and nonlinear torques. These factors present

challenges for achieving the pHRI control. To ensure compliant interaction, the characteristics of direct-drive manipulators must be considered, and the unmodeled dynamics must be compensated.

### C. Problem Description

To enable a direct-drive manipulator to interact with a human operator physically, it must be capable of compliant motion. Compliance requires the manipulator to respond to interaction forces, thereby preventing excessive forces that could cause discomfort or injury. However, direct-drive manipulators typically rely on position control and are not equipped with force sensors, which makes direct force regulation infeasible. In such cases, the interaction force must be estimated indirectly from motion states and transformed into an equivalent displacement adjustment for control.

Because the joints of direct-drive manipulators are rigidly connected to the motor rotors, the system is highly sensitive to external disturbances, and trajectory tracking accuracy is easily degraded by model uncertainties. Accordingly, the compliance problem considered in this paper can be formulated as follows: in the absence of force sensors, the interaction force should be estimated from motion states, converted into a corrected end-effector trajectory, and combined with compensation for model uncertainties to ensure accurate tracking. Through this process, the manipulator can follow the corrected trajectory precisely, thereby exhibiting compliant motion consistent with the operator's intent.

## III. FORCE SENSORLESS ADMITTANCE CONTROL OF DIRECT-DRIVE MANIPULATORS FOR PHRI

### A. Generalized momentum observer for interaction external force estimation

For the achievement of robust pHRI within the task domain, it is necessary to obtain the external interactive force  $F_{ext}$ . In direct-drive manipulators without built-in force/torque sensors, direct measurement of  $F_{ext}$  is not possible. To overcome this limitation, a force-sensorless strategy is employed, in which the generalized momentum observer (GMO) is utilized to estimate  $F_{ext}$  without relying on physical sensors.

The generalized momentum corresponding to the manipulator is defined as referenced in [16]:

$$P = M(q)\dot{q} \quad (15)$$

Differentiating (15) yields

$$\dot{P} = \dot{M}(q)\dot{q} + M(q)\ddot{q} \quad (16)$$

The manipulator dynamics satisfy the following properties: the inertia matrix  $M(q)$  is symmetric and positive definite; the matrix  $M(q) - 2C(q, \dot{q})$  is skew-symmetric [17]. According to this, (16) can be expressed as

$$\dot{P} = \bar{\tau} + \tau_{ext} \quad (17)$$

where,  $\bar{\tau} = \tau + C^T(q, \dot{q})\dot{q} - G(q) - \tau_f$ .

Let the estimated external torque be defined as

$$\hat{\tau}_{ext} = Ke \quad (18)$$

with  $e = P - \hat{P}$ . Substituting into (17) gives

$$\hat{\dot{P}} = \bar{\tau} + \hat{\tau}_{ext} \quad (19)$$

Comparing (17) and (19) leads to

$$\tau_{ext} = \dot{e} + Ke \quad (20)$$

Taking the Laplace transform of (20) and combining with (18), the following results can be obtained:

$$\hat{\tau}_{ext} = K(sI_n + K)^{-1} \tau_{ext} \quad (21)$$

which implies that for a positive definite diagonal matrix  $K$ , the estimate  $\hat{\tau}_{ext}$  converges to the actual torque  $\tau_{ext}$  as  $K \rightarrow \infty$ .

Finally, substituting (21) into (12), the estimated external force is obtained as

$$\hat{F}_{ext} = J^{+T}(q) \hat{\tau}_{ext} \quad (22)$$

#### B. Force sensorless admittance control

The block diagram of pHRI based on admittance control is shown in Fig. 4. The manipulator initially follows the planned trajectory  $x_d$ . when the operator applies an external interaction force  $F_{ext}$  at the end-effector, the admittance controller generates a corrected reference trajectory  $x_r$ . By tracking  $x_r$ , the manipulator achieves compliant behavior relative to the interaction force and adapts to the operator's movement. The admittance controller can be described as:

$$M_d(\ddot{x}_d - \ddot{x}_r) + D_d(\dot{x}_d - \dot{x}_r) + K_d(x_d - x_r) = F_{ext} \quad (23)$$

where  $M_d, D_d, K_d \in \mathbb{R}^{m \times m}$  denote the system's inertia, damping, and stiffness matrices, respectively. The desired trajectory in task space is given by  $x_d, \dot{x}_d, \ddot{x}_d \in \mathbb{R}^m$ , while the reference trajectory is described by  $x_r, \dot{x}_r, \ddot{x}_r \in \mathbb{R}^m$ . When  $M_d, D_d, K_d$  and  $x_d$  are specified, and  $F_{ext}$  is measured in real time, the corresponding reference trajectory  $x_r$  can be determined.

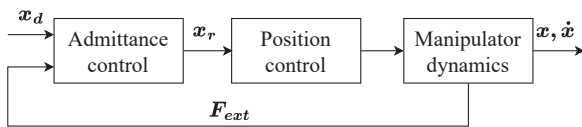


Fig. 4. Block diagram of admittance control for pHRI with force sensors

A proportional-derivative (PD) controller can be configured to realize accurate tracking of the reference trajectory  $x_r$ . However, in the absence of force-sensing devices, direct measurement of the interaction force  $F_{ext}$  is not feasible. Moreover, because direct-drive manipulators eliminate reducers,

their joint motors are more susceptible to unmodeled effects such as variable inertia, coupling, and nonlinear torques. For the convenience of analysis, it is assumed in this paper that the dynamic model parameters of the  $n$ -DOF direct-drive manipulator acquired through identification can be represented as:

$$F = \bar{M}_x(q)\ddot{x} + \bar{C}_x(q, \dot{q})\dot{x} + \bar{G}_x(q) + \bar{F}_{fx} + N(q, \dot{q}, \ddot{q}) \quad (24)$$

where,  $\bar{M}_x(q)\ddot{x}$ ,  $\bar{C}_x(q, \dot{q})\dot{x}$ , and  $\bar{G}_x(q)$  are the identified inertia, Coriolis, and gravitational terms, respectively.  $N(q, \dot{q}, \ddot{q})$  is the model uncertainty including unmodeled dynamics and identification errors.

To enhance trajectory tracking performance, this study employs a radial basis function neural network (RBFNN) to compensate for uncertainties in the dynamic model  $N(q, \dot{q}, \ddot{q})$ . The RBFNN is structured with three layers: input, hidden, and output. As a neural network with linear parameterization, its formulation can be expressed as follows [18]:

$$Y_i(Z) = W_i^T S_i(Z), \quad i = 1, 2, \dots, \nu \quad (25)$$

where,  $Z = [z_1, z_2, \dots, z_p]$  is the  $p$ -dimension input vector. The weight vector takes the form  $W_i = [w_1, w_2, \dots, w_l]^T$ , while the radial basis function is given by  $S_i(Z) = [s_1(Z), s_2(Z), \dots, s_l(Z)]^T$ . The parameter  $\nu$  refers to the total count of RBF neural networks.

The Gaussian function is adopted as the radial basis function:

$$s_j(Z) = \exp \left[ \frac{-(Z - o_j)^T (Z - o_j)}{\zeta_j^2} \right], \quad j = 1, 2, \dots, l \quad (26)$$

where,  $o_j = [o_{j1}, o_{j2}, \dots, o_{jp}]^T$  stands for the center of the receptive field, while  $\zeta$  specifies the Gaussian function's width parameter.

Given that an optimal weight  $w_i^*$  exists, the subsequent relationship can be established:

$$Y_i(Z) = W_i^{*T} S_i(Z) + \epsilon_i \quad (27)$$

where,  $\epsilon_i$  denotes the approximation error. To facilitate analysis, we introduce the term  $W_i^*$ , which is defined as the value of  $W_i$  which results in the minimum of  $|\epsilon_i|$ .

$$W_i^* = \arg \min_{W_i \in R^l} \left\{ \sup_{Z \in \Omega_z} |\epsilon_i| \right\} \quad (28)$$

The output weights  $W_i$  are updated via gradient descent according to the following rule:

$$W_i = W_i(t-1) + \eta (Y_i - \hat{Y}_i) S_i(Z) \quad (29)$$

where,  $\eta$  corresponds to the learning rate and  $\hat{Y}_i$  is the estimated output of the RBFNN. An overly high  $\eta$  can cause instability, while an overly low value slows the convergence speed.



The block diagram corresponding to the sensorless admittance control of the direct-drive manipulator is illustrated in 5. In Fig. 5, the position controller is designed as:

$$F = \bar{M}_x(q) [\ddot{x}_r + K_D(\dot{x}_r - \dot{x}) + K_P(x_r - x)] + \bar{C}_x(q, \dot{q})\dot{x} + \bar{G}_x(q) + \hat{N}(q, \dot{q}, \ddot{q}) \quad (30)$$

where  $\hat{N}(q, \dot{q}, \ddot{q})$  is defined as the total disturbance estimated via the RBFNN.  $K_P, K_D \in \mathbb{R}^{n \times n}$  denote the proportional and derivative gain matrices of the controller, respectively.

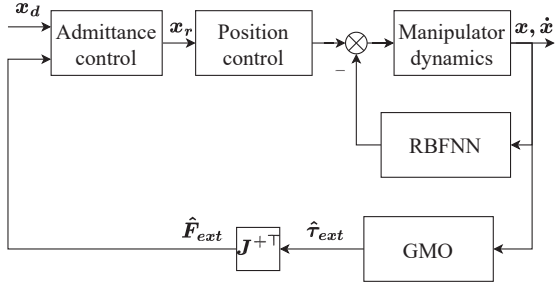


Fig. 5. Block diagram of force sensorless admittance control for pHRI with direct-drive manipulators

Upon substituting (30) into (24), the dynamics of the closed-loop system can be formulated as:

$$(\ddot{x}_r - \ddot{x}) + K_D(\dot{x}_r - \dot{x}) + K_P(x_r - x) = 0 \quad (31)$$

With suitable choices of  $K_P$  and  $K_D$ , the tracking error converges to zero, allowing the system to stably follow the desired reference trajectory.

#### IV. SIMULATION EXPERIMENT AND DISCUSSION

##### A. Simulation Setup

For the purpose of demonstrating the validity and feasibility of the presented control method, numerical simulations of pHRI are performed on a 6-DOF direct-drive manipulator (see Fig. 6 for the manipulator configuration).

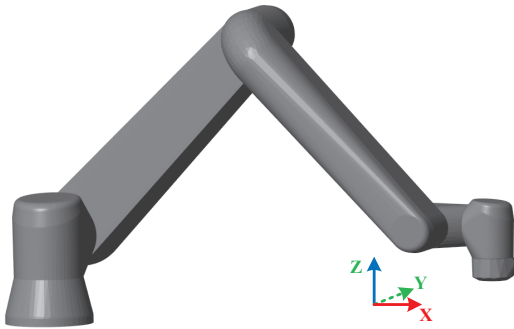


Fig. 6. The six-DOF direct drive manipulator for pHRI task

The pHRI scenario is designed as follows: from 0 to 1.5 s, the manipulator follows the planned desired trajectory.

Between 1.5 and 2.5 s, an external force of  $F_{ext} = -30$  N is applied by the operator in the Y direction, during which the manipulator complies with the operator's force. After 2.5 s, the force is removed, and the manipulator continues to execute its original task.

To better approximate the manipulator's real operating conditions in simulation, nonlinear friction disturbances are introduced at each joint. The joint friction is modeled as  $\tau = f_v \dot{q} + f_c \text{sign}(\dot{q})$ . The viscous friction coefficients  $f_v$  are set to  $[0.1, 0.1, 0.1, 0.01, 0.01, 0.01]$ , and the Coulomb friction coefficients  $f_c$  are set to  $[0.1, 0.1, 0.1, 0.01, 0.01, 0.01]$ . The function sign returns  $-1$  when  $\dot{q} < 0$  and  $+1$  when  $\dot{q} > 0$ .

In the pHRI scenario, the inertia parameters of the admittance controller are all set to 1, and the damping and stiffness parameters are set to 10 and 30, respectively. After repeated testing, the RBFNN is configured with 7 hidden-layer nodes. The initial weight is set to zero, the node centers are assigned as  $[-1.5, -1, -0.5, 0, 0.5, 1, 1.5]$ , and the width is set to 3. The positive definite diagonal matrix  $K$  of GMO is defined as  $K = \text{diag}[100, 100, 100, 100, 100, 100]$ . For the PD controller, the derivative and proportional gains are chosen as  $K_D = \text{diag}[50, 50, 50, 30, 30, 30]$  and  $K_P = \text{diag}[100, 100, 100, 40, 40, 40]$ .

##### B. Simulation results and discussion

Within the preset interaction scenario, Fig. 7 presents the GMO-estimated external force in the Y-direction, while Fig. 8 presents the trajectory tracking results of the sensorless admittance control for the direct-drive manipulator in that direction. Since interaction occurs only between 1.5–2.5 s, Fig. 7 indicates that external forces are estimated during this interval, whereas the estimates remain nearly zero at other times. As shown in Fig. 8, when the interaction force is applied by the operator,  $x_r$  adjusts correspondingly in response to the force estimated by the GMO. This demonstrates that the proposed admittance controller can generate reference trajectories consistent with the operator's intent. Furthermore, the tracking results confirm that the end effector of the direct-drive manipulator can accurately track the reference trajectory. Based on the results, we can conclude that the proposed controller not only ensures precise trajectory tracking but also enables compliant adaptation to the operator's motion.

#### V. CONCLUSION

This paper puts forward a force-free sensorless admittance control method aiming to realize the compliant interaction of a direct-drive manipulator. By employing the GMO for interaction force estimation and the RBFNN for uncertainty compensation, the proposed approach achieves reliable compliant behavior even in position control mode. Results from simulations on a six-DOF direct-drive manipulator showed that the proposed method not only realizes precise trajectory tracking but also effectively accommodates operator motion, leading to improved safety and compliance in human-robot interaction.

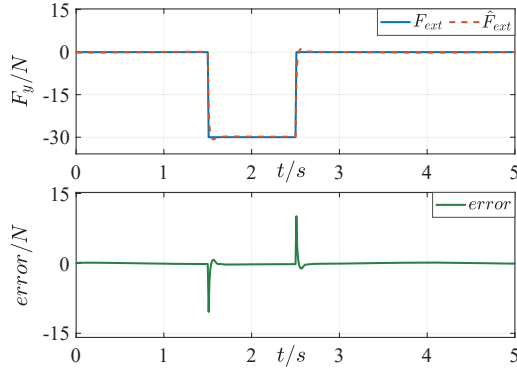


Fig. 7. Interaction external forces estimation results based on GMO

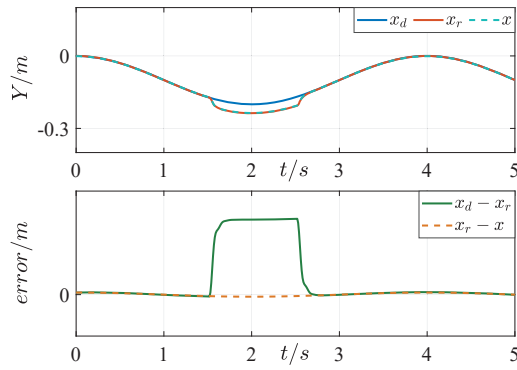


Fig. 8. Trajectory tracking results of the end-effector in the Y direction

Although the simulation results confirm its feasibility, experimental validation on a physical direct-drive manipulator platform will be the focus of future work to further verify its effectiveness and robustness. Moreover, developing adaptive admittance control algorithms for complex and dynamic interaction scenarios, and leveraging the advantages of direct-drive manipulators in fast response and high-precision positioning to design a sensorless control framework that combines compliance with accuracy, remain important directions for further research.

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